

**CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY**  
**II and IV Semester of BSc Physics Examination May 2018**  
**PD 155 Chemistry – II**

**Date: 03/05/2018**

**Day: Thursday**

**Time: 01.30 pm to 02.00 pm**

**Maximum Marks: 20**

**MCQ**

**Important Instructions:**

- Tick the correct answer and it should be written in question paper itself.
- Use of non-programmable calculator is allowed.

**Que:1 Answer the following multiple choice questions.**

**[20]**

- 1** Phenol reacts with bromine water to give the precipitate of \_\_\_\_\_  
(a) Bromophenol (b) Dibromophenol  
(c) Tribromophenol (d) None of these
- 2** Which of the following reagents cannot be used to prepare an alkyl chloride from an alcohol?  
(a)  $\text{HCl} + \text{ZnCl}_2$  (b)  $\text{SOCl}_2$   
(c)  $\text{NaCl}$  (d)  $\text{PCl}_5$
- 3** The reaction of ethyl bromide with sodium ethoxide give \_\_\_\_\_  
(a) Ethyl methyl ether (b) Diethyl ether  
(c) Dimethyl ether (d) Above all
- 4** Haloform reaction is diagnostic test for \_\_\_\_\_  
(a)  $-\text{CHO}$  (b)  $-\text{C}=\text{O}$  (c)  $-\text{COCH}_3$  (d) all of these
- 5** Iodobenzene heated with copper to form \_\_\_\_\_  
(a) Biphenyl (b) Biiodophenyl  
(c) Trphenyl (d) None of these
- 6** Ethers react with cold  $\text{H}_2\text{SO}_4$  to form \_\_\_\_\_  
(a) Alkenes (b) Zwitterions  
(c) Alkoxides (d) Oxonium salts
- 7** Lucas test is used to determine the type of \_\_\_\_\_  
(a) Acids (b) Alcohols (c) Ketones (d) Esters
- 8** Hydroboration-oxidation of propene gives \_\_\_\_\_  
(a) Isopropyl alcohol (b) n-propyl alcohol  
(c) Isobutyl alcohol (d) Tert-Butyl alcohol

- 9 Anisole is formed when phenol is treated with \_\_\_\_\_  
 (a)  $\text{CH}_3\text{I}/\text{NaOH}$  (b)  $\text{CH}_3\text{CH}_2\text{I}/\text{NaOH}$   
 (c)  $\text{CHCl}_3/\text{NaOH}$  (d) Acetic anhydride
- 10 Acetone reacts with HCN to form cyanohydrins. It is an example of \_\_\_\_\_  
 (a) Electrophilic addition (b) Electrophilic substitution  
 (c) Nucleophilic addition (d) None of these
- 11 The study of the flow of heat or any other form of energy into or out of a system undergoing physical or chemical change is called \_\_\_\_\_  
 (a) Thermochemistry (b) Thermo kinetics  
 (c) Thermodynamics (d) Thermochemical studies
- 12 A system that can transfer neither matter nor energy to and from its surroundings is called \_\_\_\_\_  
 (a) A closed system (b) An isolated system  
 (c) An open system (d) A homogeneous system
- 13 Zinc granules reacting with dilute hydrochloric acid in an open beaker constitutes \_\_\_\_\_  
 (a) An isolated system (b) An open system  
 (c) A closed system (d) A heterogeneous system
- 14 A system in which state variables have constant values throughout the system is called in a state of \_\_\_\_\_  
 (a) Equilibrium (b) Non-equilibrium  
 (c) Isothermal equilibrium (d) None of these
- 15 Which of the following relations is applicable to adiabatic expansion of an ideal gas?  
 (a)  $T_1 V_1^{\gamma-1} = T_2 V_2^{\gamma-1}$  (b)  $P_1 V_1^{\gamma} = P_2 V_2^{\gamma}$  (c)  $P_1 V_1 = P_2 V_2$  (d) none of these
- 16 When a salt is added to a solution of another salt having a common ion, the degree of dissociation  $\alpha$  \_\_\_\_\_  
 (a) Increases (b) Remain same  
 (c) Decreases (d) All of above
- 17 Ostwald's dilution law is applicable to \_\_\_\_\_  
 (a) All electrolytes (b) Strong electrolytes  
 (c) Weak electrolytes (d) None of these
- 18 A weak acid together with a salt of the same acid with a strong base. These are called \_\_\_\_\_  
 (a) Acidic buffer (b) Basic buffer  
 (c) Both (d) None of these
- 19 The fraction of the amount of the electrolyte in the solution present as a free radical is called \_\_\_\_\_  
 (a) Dissociation (b) Precipitation  
 (c) Degree of dissociation (d) Degree of precipitation
- 20 The degree of dissociation of an electrolyte depends upon \_\_\_\_\_  
 (a) Nature of solute (b) Nature of solvent  
 (c) Concentration of solute (d) All of these

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**Date: 03/05/2018    Day: Thursday    Time: 02.00 pm to 04.30 pm    Maximum Marks: 50**

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**Important Instructions:**

- **Section I and II must be attempted in TWO ANSWER SHEET.**
- Make suitable assumptions and draw neat figures wherever required.
- Use of non-programmable calculator is allowed.
- Show necessary calculations.

**SECTION -I**

**[20]**

**Q - II**

- [A]** Give the synthesis of crotonaldehyde and mesityl oxide. **[04]**

**OR**

- [A]** Write a note on: Reimer-Tiemann reaction. **[04]**

- [B]** Write a note on: Cannizzaro reaction. **[03]**

- [C]** Write a note on: Reformatsky reaction. **[03]**

**Q - III**

- [A]** Write a note on: Kolbe reaction. **[04]**

- [B]** Why the aryl halides are less reactive towards the nucleophilic displacement reaction? **[03]**

- [C]** Give the mechanism of sulphonation of benzene. **[03]**

**OR**

- [C]** Give the synthesis of aldehyde and ketone from the geminal dihalides. **[03]**

**SECTION -II**

**[30]**

**Q - IV**

- [A]** Define heat capacity at constant pressure ( $C_p$ ), heat capacity at constant volume ( $C_v$ ) and derive the relation  $C_p - C_v = R$ . **[04]**

**OR**

- [A]** Derive the equation  $P_1 V_1^\gamma = P_2 V_2^\gamma$  for adiabatic expansion of an ideal gas. **[04]**

- [B]** Explain the Kirchhoff's law in detail. **[03]**

- [C]** Explain common-ion effect by giving suitable examples. **[03]**

**Q - V**

- [A]** Write two preparation methods for phenol. **[04]**

- [B]** Write the oxidation reaction of 1°, 2°, 3° alcohol. **[03]**

- [C]** Explain the Williamson ether synthesis by giving suitable example. **[03]**

**OR**

- [C]** Write the Friedel-Crafts alkylation reaction and explain its mechanism. **[03]**

**Q - VI**

- [A]** Explain the Debye-Huckel theory. **[04]**

- [B]** Derive the Henderson-Hasselbalch equation for buffer solution. **[03]**

- [C]** Write the factors which influence the degree of dissociation. **[03]**

**OR**

- [C]** Calculate the value of  $\Delta E$  and  $\Delta H$  on heating 64.0 g of oxygen from 0 °C to 100°C.  $C_v$  and  $C_p$  on an average are 5.0 and 7.0 cal mol<sup>-1</sup> degree<sup>-1</sup>. **[03]**